

Three phase current --- Solution

X23 (2023) — English translation

A1 The line voltage between terminals a and b can be represented as $\mathcal{E}_{ab} = \mathcal{E}'_0 \cos(\omega t - \Delta\varphi)$, $\mathcal{E}'_0 > 0$. Find $\mathcal{E}'_0$ and $\Delta\varphi$. Express your answer using $\mathcal{E}_0$. **0.60**

$$\mathcal{E}_{ab} = \mathcal{E}_a - \mathcal{E}_b = \mathcal{E}_0 \left(\cos(\omega t) - \cos\left(\omega t - \frac{2\pi}{3}\right) \right).$$

Применив формулу разности косинусов, получим:

$$\mathcal{E}_{ab} = 2\mathcal{E}_0 \left(\sin\left(\omega t - \frac{\pi}{3}\right) \sin\left(-\frac{\pi}{3}\right) \right) = -\sqrt{3}\mathcal{E}_0 \sin\left(\omega t - \frac{\pi}{3}\right).$$

Заменяем синус на требуемый косинус:

$$\mathcal{E}_{ab} = \sqrt{3}\mathcal{E}_0 \cos\left(\omega t + \frac{\pi}{6}\right).$$

Note. В условии не была указана полярность $\mathcal{E}_{ab}$, поэтому возможен вариант $\mathcal{E}_{ab} = \mathcal{E}_b - \mathcal{E}_a$. В этом случае знак косинуса в окончательно выражении поменяется, что соответствует сдвигу фазы косинуса на π :

$$\mathcal{E}_{ab} = \sqrt{3}\mathcal{E}_0 \cos\left(\omega t - \frac{5\pi}{6}\right).$$

Answer:

$$\mathcal{E}'_0 = \sqrt{3}\mathcal{E}_0 \Delta\varphi = -\frac{\pi}{6} \quad \text{либо} \quad \Delta\varphi = \frac{5\pi}{6}$$

A2 Let each of the load elements (Z_a, Z_b, Z_c) be a resistor $R = 9 \text{ кОм}$ and a capacitor $C = 2 \cdot 10^{-7} \text{ Ф}$ connected in series. Find the current amplitudes through the line wire I_a and the neutral wire I_0 (formula and numerical value). **1.00**

Let's use the method of complex amplitudes. We will denote complex amplitudes by a line over the corresponding quantities. Let's write Kirchhoff's 2nd law for a circuit with current I_a :

$$\mathcal{E}_a = \overline{I}_a(R + r) - \overline{I}_a \frac{i}{\omega C} + \overline{I}_0 r; \quad (1)$$

Аналогично для I_b, I_c :

$$\mathcal{E}_b = \overline{I}_b(R + r) - \overline{I}_b \frac{i}{\omega C} + \overline{I}_0 r; \quad (2)$$

$$\mathcal{E}_c = \overline{I}_c(R + r) - \overline{I}_c \frac{i}{\omega C} + \overline{I}_0 r. \quad (3)$$

Также запишем 1-й закон Кирхгофа:

$$\overline{I}_0 = \overline{I}_a + \overline{I}_b + \overline{I}_c. \quad (4)$$

Выразим $\overline{I}_a, \overline{I}_b, \overline{I}_c$ из (1), (2), (3) соответственно и подставим в (4):

$$\overline{I}_0 = \frac{\mathcal{E}_a - \overline{I}_0 r}{R + r - \frac{i}{\omega C}} + \frac{\mathcal{E}_b - \overline{I}_0 r}{R + r - \frac{i}{\omega C}} + \frac{\mathcal{E}_c - \overline{I}_0 r}{R + r - \frac{i}{\omega C}}. \quad \text{[19]} \quad \mathcal{E}_a + \mathcal{E}_b + \mathcal{E}_c = 0,$$

получим

$$\overline{I}_0 = 0.$$

С учетом этого имеем ответ для $\overline{I}_a$:

$$\overline{I}_a = \frac{\mathcal{E}_0}{(R + r) - \frac{i}{\omega C}}, |\overline{I}_a| = I_a = \frac{\mathcal{E}_0}{\sqrt{(R + r)^2 + \frac{1}{(\omega C)^2}}}.$$

Answer:

$$I_a = \frac{\mathcal{E}_0}{\sqrt{(R + r)^2 + \frac{1}{(\omega C)^2}}} \approx 14,6 \text{ мА};$$

$$I_0 = 0.$$

A3 Find the efficiency of the circuit from point A2 η (formula and numerical value). Consider the efficiency factor to be the ratio of the average power released at the load to the average power released in the entire circuit. 0.40

The power in the circuit is allocated only through resistors. Useful power is released on resistors R . Let the phase shift for this resistor be $\Delta\varphi_1$. Power across resistor a :

$$P_a = |\overline{I_a}|^2 \cos^2(\omega t - \Delta\varphi_1) R = \frac{1}{2} |\overline{I_a}|^2 R (1 + \cos(2(\omega t - \Delta\varphi_1))).$$

Так как среднее $\cos(2(\omega t - \Delta\varphi_1))$ за период равно 0, средняя за период мощность:

$$\langle P_a \rangle = \frac{1}{2} |\overline{I_a}|^2 R.$$

Видно, что сдвиг фаз $\Delta\varphi_1$ не влияет на $\langle P_a \rangle$, а значит:

$$\langle P_a \rangle = \langle P_b \rangle = \langle P_c \rangle.$$

Итого полезная мощность:

$$P_{пол} = \frac{3}{2} |\overline{I_a}|^2 R.$$

Аналогично находим общую мощность в цепи:

$$P_{вся} = \frac{3}{2} |\overline{I_a}|^2 (R + r).$$

По определению, данному в условии:

$$\eta = \frac{P_{пол}}{P_{вся}} = \frac{R}{R + r} \approx 64,3\%.$$

Answer:

$$\eta = \frac{R}{R + r} \approx 64,3\%.$$

A4 Now let the load be asymmetrical: elements Z_b and Z_c are resistors with resistance R , and element Z_a is a capacitor with capacitance $C_1 = 10^{-9}$ Ф. Find the numerical value of the efficiency of circuit η_1 in this case. It is not necessary to obtain an analytical expression for η_1 . Use the numerical data from point A2. 1.00

Capacitor impedance module:

$$|Z_a| = \frac{1}{\omega C} = \frac{1}{2\pi f C} \approx 3183 \text{ Ом}.$$

Заметим, что $|Z_a| \gg R$, поэтому при расчете токов током через конденсатор можно пренебречь. Аналогично пункту A2 напомним законы Кирхгофа для цепи:

$$\overline{I_0} = \overline{I_b} + \overline{I_c}; \quad (1)$$

$$\mathcal{E}_0 \left(-\frac{\sqrt{3}}{2} i - \frac{1}{2} \right) = \overline{I_b}(R + r) + \overline{I_0}r; \quad (2)$$

$$\mathcal{E}_0 \left(\frac{\sqrt{3}}{2} i - \frac{1}{2} \right) = \overline{I_c}(R + r) + \overline{I_0}r. \quad (3)$$

Сложим (2) и (3) [11](1). Таким образом найдем $\overline{I_0}$:

$$\overline{I_0} = -\frac{\mathcal{E}_0}{R + 3r}.$$

Подставив найденное $\overline{I_0}$ в (2) и (3), находим $\overline{I_b}$ и $\overline{I_c}$:

$$\overline{I_b} = \frac{\mathcal{E}_0}{R + r} \left(-\frac{\sqrt{3}}{2} i + \left(\frac{r}{R + 3r} - \frac{1}{2} \right) \right); \quad \overline{I_c} = \frac{\mathcal{E}_0}{R + r} \left(\frac{\sqrt{3}}{2} i + \left(\frac{r}{R + 3r} - \frac{1}{2} \right) \right);$$

Выражения для модулей:

$$|\overline{I_0}| = \frac{\mathcal{E}_0}{R + 3r} \approx 12,92 \text{ mA}; \quad |\overline{I_b}| = |\overline{I_c}| = \frac{\mathcal{E}_0}{R + r} \sqrt{\frac{3}{4} + \left(\frac{r}{R + 3r} - \frac{1}{2} \right)^2} \approx 20,23 \text{ mA};$$

Находим средние полезную и общую мощности:

$$P_{пол} = 2 \frac{1}{2} |\overline{I_b}|^2 R; \quad P_{вся} = 2 \frac{1}{2} |\overline{I_b}|^2 (R + r) + \frac{1}{2} |\overline{I_0}|^2 r; \quad \eta_1 = \frac{2 |\overline{I_b}|^2 R}{2 |\overline{I_b}|^2 (R + r) + |\overline{I_0}|^2 r} \approx 59,9\%.$$

Answer:

$$\eta_1 \approx 59,9\%.$$

- B1** Let a current I_0 flow in one of the square frames with side a at a given moment of time. Find the magnetic field B_0 at the center of the square created only by this frame. **0.60**

First, consider a segment with current I . Let us find the field at a point located at a distance h from the line containing it. Biot-Savart-Laplace law:

$$d\vec{B} = \frac{\mu_0 I_0}{4\pi} \frac{[\vec{dl} \times \vec{r}]}{|\vec{r}|^3}.$$

Поле от любого малого фрагмента отрезка поле будет направлено одинаково – по Oz . С учетом этого запишем выражение для проекции поля $d\vec{B}$ на Oz :

$$dB_z = \frac{\mu_0 I_0}{4\pi} \frac{dl \cos(\alpha)}{r^2}.$$

Подставим $dl = \frac{h d\alpha}{\cos^2(\alpha)}$, $r = \frac{h}{\cos(\alpha)}$:

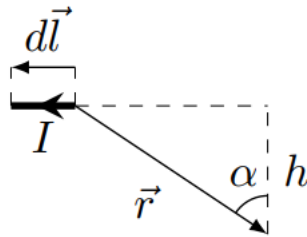
$$dB_z = \frac{\mu_0 I_0}{4\pi h} \cos(\alpha) d\alpha; B_z = \frac{\mu_0 I_0}{4\pi h} \int_{\alpha_1}^{\alpha_2} \cos(\alpha) d\alpha; B_z = \frac{\mu_0 I_0}{4\pi h} (\sin(\alpha_2) - \sin(\alpha_1)).$$

For one side of the square frame $h = a/2$, $\alpha_2 = -\alpha_1 = \pi/4$. Then the margin in the center of the frame:

$$B_0 = 4 \frac{\mu_0 I_0}{4\pi \frac{a}{2}} \left(2 \frac{\sqrt{2}}{2} \right) = 2\sqrt{2} \frac{\mu_0 I_0}{\pi a}.$$

Answer:

$$B_0 = 2\sqrt{2} \frac{\mu_0 I_0}{\pi a}.$$



- B2** We will now consider the field that is created by all three frames, the currents through which are indicated in the introduction. Prove that in this case the magnetic field rotates with a constant angular velocity in the plane xy , that is, its modulus is constant and equal to B_1 , and the angle with the axis x $\psi = \omega t + \psi_0$ (then $B_x = B_1 \cos \psi$, $B_y = B_1 \sin \psi$). Find B_1 and ψ_0 . Express your answer using B_0 . **1.50**

We obtain the projection of the field from the three frames onto Ox :

$$B_x = B_0 \cos(\omega t) + B_0 \cos\left(\omega t - \frac{2\pi}{3}\right) \cos\left(\frac{2\pi}{3}\right) + B_0 \cos\left(\omega t - \frac{4\pi}{3}\right) \cos\left(\frac{4\pi}{3}\right) = B_0 \left(\cos(\omega t) - \frac{1}{2} \left(\cos\left(\omega t - \frac{2\pi}{3}\right) + \cos\left(\omega t - \frac{4\pi}{3}\right) \right) \right).$$

Применим формулу суммы косинусов:

$$B_0 \left(\cos(\omega t) - \cos(\omega t - \pi) \cos\left(\frac{\pi}{3}\right) \right) = \frac{3}{2} B_0 \cos(\omega t).$$

Аналогично можно рассчитать проекцию суммарного поля на Oy :

$$B_y = B_0 \cos\left(\omega t - \frac{2\pi}{3}\right) \sin\left(\frac{\pi}{3}\right) - B_0 \cos\left(\omega t - \frac{4\pi}{3}\right) \sin\left(\frac{\pi}{3}\right), B_y = \frac{\sqrt{3}}{2} 2 \sin(\omega t - \pi) \sin\left(-\frac{\pi}{3}\right) = \frac{3}{2} B_0 \sin(\omega t).$$

Оба выражения приводят к ответу:

$$B_1 = \frac{3}{2} B_0; \quad \psi_0 = 0.$$

Альтернативное решение. Let $\vec{e}_x$, $\vec{e}_y$ be the unit vectors of the x , y axes, respectively. Let us use them to write down vector expressions for fields from three frames:

$$\vec{B}_a = B_0 \vec{e}_x \cos(\omega t); \vec{B}_b = B_0 \left(-\frac{1}{2} \vec{e}_x + \frac{\sqrt{3}}{2} \vec{e}_y \right) \cos \left(\omega t - \frac{2\pi}{3} \right); \vec{B}_c = B_0 \left(-\frac{1}{2} \vec{e}_x - \frac{\sqrt{3}}{2} \vec{e}_y \right) \cos \left(\omega t - \frac{4\pi}{3} \right).$$

Перейдем к комплексным амплитудам:

$$\begin{aligned} \vec{B}_a &= B_0 \vec{e}_x e^{i\omega t}; \vec{B}_b = B_0 \left(-\frac{1}{2} \vec{e}_x + \frac{\sqrt{3}}{2} \vec{e}_y \right) e^{i\omega t} e^{-i\frac{2\pi}{3}}; \vec{B}_c = B_0 \left(-\frac{1}{2} \vec{e}_x - \frac{\sqrt{3}}{2} \vec{e}_y \right) e^{i\omega t} e^{-i\frac{4\pi}{3}} = B_0 \left(-\frac{1}{2} \vec{e}_x - \frac{\sqrt{3}}{2} \vec{e}_y \right) e^{i\omega t} e^{i\frac{2\pi}{3}}. \\ \vec{B}_1 &= \vec{B}_a + \vec{B}_b + \vec{B}_c = B_0 e^{i\omega t} \left(\vec{e}_x \left(1 - \frac{e^{i\frac{2\pi}{3}} + e^{-i\frac{2\pi}{3}}}{2} \right) - \sqrt{3} i \vec{e}_y \frac{e^{i\frac{2\pi}{3}} - e^{-i\frac{2\pi}{3}}}{2i} \right) == B_0 e^{i\omega t} \left(\vec{e}_x \left(1 - \cos \left(\frac{2\pi}{3} \right) \right) - \sqrt{3} i \vec{e}_y \sin \left(\frac{2\pi}{3} \right) \right) == B_0 e^{i\omega t} \end{aligned}$$

Взяв действительную часть от выражения выше, получим:

$$\vec{B}_1 = \frac{3}{2} B_0 (\vec{e}_x \cos(\omega t) + \vec{e}_y \sin(\omega t)).$$

This implies the answer obtained earlier.

Answer:

$$B_1 = \frac{3}{2} B_0; \quad \psi_0 = 0.$$

B3 The small box in the center is rotated so that its normal makes an angle α with the axis x . Find the magnetic flux through it at time t . Express your answer using B_1 , S , α , ω . **0.40**

Answer:

$$\Phi = B_1 S \cos(\alpha - \omega t).$$

B4 Let the inductance of the frame be L_2 and the resistance R_2 . The frame rotates at a constant angular velocity ω_1 , so $\alpha = \omega_1 t$ ($0 < \omega_1 < \omega$). In the steady state, the current in the loop depends on time as **1.00**

$$I(t) = A \cos(\omega_2 t - \varphi_2).$$

Найдите A , ω_2 и φ_2 . Выразите ответ через параметры рамки, B_1 , ω , ω_1 .

The total magnetic field flux through the frame is created by the rotating field and current in the frame I

$$\Phi_t = \Phi + L_2 I$$

Суммарная ЭДС индукции в рамке с током I :

$$\mathcal{E} = \dot{\Phi}_t = -\dot{\Phi} - L_2 \dot{I}.$$

Из закона Ома для рамки $IR_2 = \mathcal{E}$ получим

$$-\dot{\Phi} = L_2 \dot{I} + IR_2$$

Это уравнение формально эквивалентно уравнению для RL -цепи с источником переменного напряжения $-\dot{\Phi}$. Будем записывать колебания в комплексном виде:

$$\Phi = B_1 S e^{i(\omega - \omega_1)t}.$$

Поскольку поток меняется с частотой $\omega_2 = \omega - \omega_1$, с той же частотой будет меняться и ток. ЭДС индукции в рамке:

$$\mathcal{E}(t) = -\dot{\Phi} = -i\omega_2 B_1 S e^{i\omega_2 t}.$$

Подставим зависимость тока от времени в виде $I = \bar{I} e^{i\omega_2 t}$, получим

$$(i\omega_2 L_2 + R_2) \bar{I} = -i\omega_2 B_1 S;$$

$$\bar{I} = -\frac{i(\omega - \omega_1) B_1 S}{i(\omega - \omega_1) L_2 + R_2}.$$

Отсюда получаем ответы

$$\omega_2 = \omega - \omega_1, A = |\bar{I}| = \frac{B_1 S (\omega - \omega_1)}{\sqrt{R_2^2 + (\omega - \omega_1)^2 L_2^2}}; \varphi_2 = \pi - \arctan \left(\frac{R_2}{(\omega - \omega_1) L_2} \right).$$

Answer:

$$\omega_2 = \omega - \omega_1; A = \frac{B_1 S(\omega - \omega_1)}{\sqrt{R_2^2 + (\omega - \omega_1)^2 L_2^2}}; \varphi_2 = \pi - \arctan\left(\frac{R_2}{(\omega - \omega_1)L_2}\right).$$

- B5** In order to ensure more uniform rotation of the rotor, instead of one frame we will use three frames, rotated relative to each other by 120° around the z axis (that is, if the position of one frame is specified by the angle α , the positions of the remaining frames are specified by the angles $\alpha_2 = \alpha + 2\pi/3$, $\alpha_3 = \alpha + 4\pi/3$). The frames are isolated from each other; their mutual induction can be neglected. Prove that the projection onto the z axis of the moment of forces acting on the structure of three frames is constant. Find its value M_z . Express your answer in terms of the frame parameters, B_1, ω, ω_1 . 1.50

The magnetic flux of the rotating magnetic field through the i th frame ($i = 1, 2, 3$) is equal to $\Phi_i = B_1 S \cos(\omega_2 t - \alpha_{0i})$ ($\alpha_{01} = 0$, $\alpha_{02} = 2\pi/3 = \Delta\alpha$, $\alpha_{03} = 4\pi/3 = 2\Delta\alpha$). Thus, the magnetic flux through the second and third frames is delayed by $2\pi/3$ and $4\pi/3$. Then the currents through these frames are delayed by the same phase. Magnetic moment of a frame with current I area S

$$m = IS,$$

на i -тую рамку действует момент сил

$$\vec{M}_i = [\vec{m}_i \times \vec{B}].$$

В проекции на Oz :

$$M_{iz} = m_i B_1 \sin(\psi - \alpha_i).$$

Таким образом (обозначения для тока взяты из предыдущего пункта):

$$M_z = B_1 A S (\sin(\omega_2 t) \cos(\omega_2 t - \varphi_2) + \sin(\omega_2 t - \Delta\alpha) \cos(\omega_2 t - \varphi_2 - \Delta\alpha) + \sin(\omega_2 t - 2\Delta\alpha) \cos(\omega_2 t - \varphi_2 - 2\Delta\alpha)).$$

Сумма трех синусов, зависящих от времени, обращается в 0, поэтому

$$M_z = \frac{3}{2} B_1 A S \sin \varphi_2.$$

Подставив φ_2, A , получим

$$\sin \varphi_2 = \sin\left(\arctan \frac{R_2}{(\omega - \omega_1)L_2}\right) = \frac{R_2}{\sqrt{R_2^2 + (\omega - \omega_1)^2 L_2^2}}.$$

Альтернативное решение. Let's find the total moment of three frames with current. This problem is mathematically equivalent to the problem of finding the total magnetic field from three external current-carrying loops. Indeed, the magnetic moment of each subsequent frame is rotated by 120° degrees relative to the magnetic moment of the previous frame, and is delayed in phase by $2\pi/3$. Let us first consider the problem in the reference frame of a rotating frame. Then if $m_0 = AS$ is the amplitude of the magnetic moment from one frame, then the value of the total magnetic moment is $m_1 = \frac{3}{2}m_0$, and the magnetic moment rotates with an angular velocity ω_2 . In this case, the initial phase of the magnetic moment is equal to $-\varphi_2$, since all currents lag relative to the magnetic field by this phase. Since the frames themselves rotate with an angular velocity of ω_1 , relative to the laboratory frame of reference, the magnetic moment rotates with an angular velocity of $\omega_1 + \omega_2 = \omega$. The angular velocities of the magnetic moment and the magnetic field coincide, so the angle between them remains constant. It follows that the moment of forces is also constant and equal to

$$M_z = \frac{3}{2} m_0 B_1 \sin \varphi_2 = \frac{3}{2} A S B_1 \sin \varphi_2.$$

Thus, we obtain the same answer.

Answer:

$$M_z = \frac{3}{2} B_1^2 S^2 \frac{(\omega - \omega_1) R_2}{(\omega - \omega_1)^2 L_2^2 + R_2^2}.$$

- B6** Construct a qualitative graph of the dependence $M_z(\omega_1)$ in the frequency range $0 < \omega_1 < \omega$. Specify the coordinates of the characteristic points. Consider that $R_2/L_2 < \omega$. 1.00

Let's rewrite the expression for the moment of forces in the form

$$M = \frac{3}{2} B_1^2 S^2 \frac{R_2}{L_2^2} \frac{1}{(\omega - \omega_1) + \frac{R_2^2}{L_2^2} \frac{1}{(\omega - \omega_1)}}.$$

Максимуму момента отвечает минимум знаменателя

$$(\omega - \omega_1) + \frac{R_2^2}{L_2^2} \frac{1}{(\omega - \omega_1)}.$$

Взяв производную по $\omega - \omega_1$, получим, что экстремуму соответствует

$$\omega - \omega_{1max} = \frac{R_2}{L_2}; \omega_{1max} = \omega - \frac{R_2}{L_2}.$$

Максимальное значение момента $M_{max} = \frac{3}{4} \frac{B_1^2 S^2}{L_2}$. Любопытно отметить, что M_{max} не зависит от R_2 . При $\omega_1 = 0$

$$M_0 = \frac{3}{2} B_1^2 S^2 \frac{\omega R_2}{\omega^2 L_2^2 + R_2^2}.$$

